

The Turbo Avenger: A Multi-Actuator Autonomous Payload Delivery System**2/52****#221****Turbo Avengers****Date:** April 6, 2026**Submitted to:** RHS Cornerstone Limited**Submitted by:**

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Executive Summary:

Turbo Avengers (Team 221) was commissioned by RHS Cornerstone Limited to develop an autonomous payload delivery system for hazardous search-and-rescue environments. The team engineered a servo-controlled elastic catapult featuring a synchronized three-actuator system for autonomous arming, loading, and firing. The prototype successfully met mission objectives by delivering exactly three payload units to a 30 × 40 cm target from a distance of 3 meters. With a total cost of \$328.00, the system provides a robust, cost-effective, and repeatable solution. Testing confirmed that the dual-servo autoloader eliminates mechanical jams; however, precise timing control is required to maintain actuator synchronization. Overall, the Turbo Avenger demonstrates a reliable and scalable solution for autonomous payload delivery in hazardous environments.

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1.0 Introduction

1.1 Problem Statement

RHS Cornerstone Limited requires the development of an autonomous system capable of delivering emergency payloads into restricted search-and-rescue environments. The system must autonomously navigate a predefined course, stop precisely at a designated launch zone, and accurately deliver exactly three payload units into a 30 × 40 cm target area from a distance of 3 meters without human intervention. The solution must demonstrate high reliability, repeatability, and compliance with all physical and operational constraints.

1.2 Requirements

Functions:

The system must autonomously navigate a predefined course, stop precisely at the designated drop zone, sequentially reload payloads without failure, and execute three controlled and consistent launches to meet mission requirements.

Objectives:

The design aims to maximize launch accuracy within ± 5 cm of the target, ensure highly repeatable and consistent performance across multiple trials, and minimize the likelihood of mechanical failure or jamming during operation.

Constraints:

The system is constrained to utilize elastic potential energy as the primary launch mechanism, fit entirely within the dimensional limits of the ELEGOO robotic chassis, and incorporate non-destructive mounting methods to preserve structural integrity and allow component reuse.

1.3 Background Research

The development of autonomous search-and-rescue systems requires the integration of reliable navigation with precise mechanical actuation. Murphy et al. (2011) emphasize that system reliability is the primary determinant of mission success, as failure in critical subsystems, specifically energy release mechanisms, which can result in total mission failure. Historically, projectile systems such as ballista-style catapults demonstrated effective methods of storing and releasing elastic potential energy. However, these systems relied heavily on human input, introducing variability in launch force, timing, and accuracy. Modern engineering solutions address these limitations through microcontroller-based automation and actuator synchronization. Smith and Jones (2022) found that replacing manual triggering mechanisms with digitally controlled systems significantly improves trajectory consistency by eliminating human-induced variability. Similarly, Lee and Patel (2019) highlight the importance of non-destructive system integration to maintain platform stability and ensure compatibility with onboard sensors and structural components. These findings had a direct impact on the design of the Turbo Avenger. The implementation of a synchronized three-actuator system, controlled by an Arduino UNO R3 and

ESP32, enables precise control of energy storage and release. The inclusion of a DRV8833 motor driver ensures accurate actuator coordination, resulting in consistent launch trajectories with an accuracy of approximately ± 5 cm over a 3-meter range.

2.0 Proposed Design Details

2.1 Mechanical Architecture: Functional Description

The Turbo Avenger utilizes a pivoting launch arm powered by elastic potential energy to deliver payloads with consistent force and trajectory. The system operates using a synchronized three-actuator control system. The overall mechanical architecture of the Turbo Avenger is shown in Figure 1

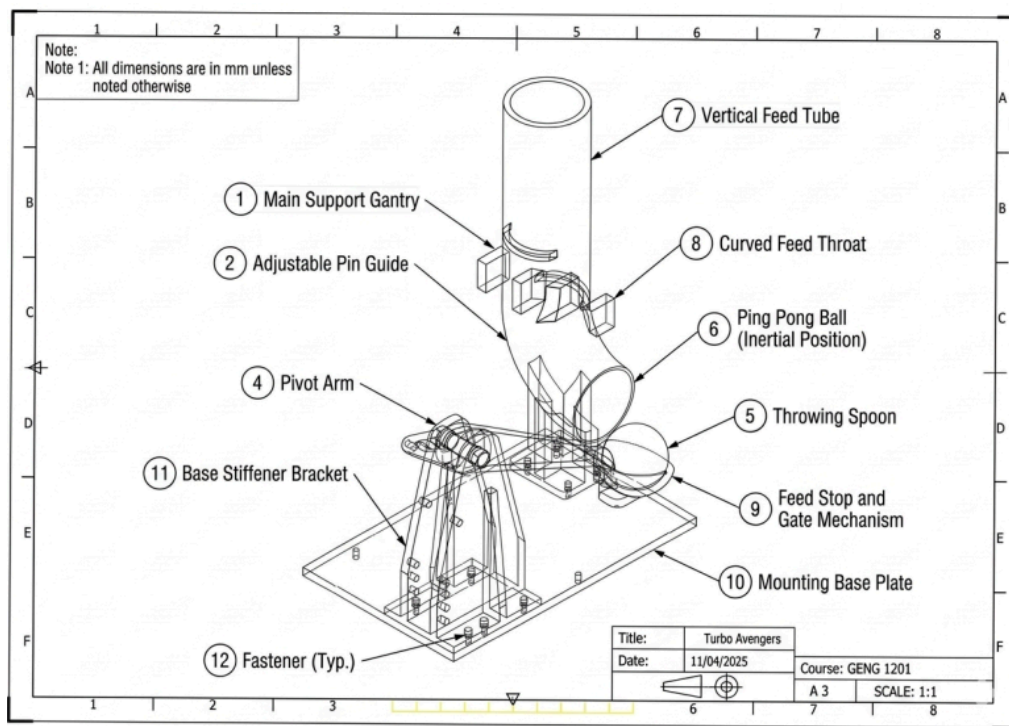


Figure 1: Mechanical architecture of the Turbo Avenger showing the pivoting launch arm, servo-driven pulley system, locking latch, and dual-servo autoloading mechanism.

The servo-driven pulley mechanism tensions the launch arm to a fixed and controlled position, directly supporting the objective of consistent trajectory output. This mechanism is powered through a DRV8833 motor driver, enabling reversible polarity for controlled autonomous arming and tension release. A locking servo latch secures the arm during loading and releases it on command, ensuring safe and controlled energy discharge. Additionally, a dual-servo autoloading mechanism delivers payloads sequentially, eliminating mechanical jamming and ensuring the system fulfills the requirement of reloading and launching exactly three payloads autonomously. This integrated design enables the system to perform all required functions, including autonomous navigation interfacing, precise stopping, sequential reloading, and controlled launching. The design

meets all constraints by utilizing elastic potential energy, maintaining compatibility within the ELEGOO chassis dimensions, and employing non-destructive mounting techniques. Furthermore, this three-actuator architecture is inherently scalable and can be adapted for higher-load, full-scale applications such as industrial or emergency payload delivery systems while maintaining control precision and system reliability. Detailed engineering drawings are provided in Appendix B.

2.2 Full-Scale Concept

The full-scale system maintains the same three-actuator architecture validated by the prototype, with scaled structural components and actuation forces to support real-world deployment. The Turbo Avenger prototype represents a scaled version of an autonomous payload delivery system intended for search-and-rescue environments such as debris fields and restricted urban zones. The full-scale design extends the same three-actuator architecture to support real-world operation with increased payload capacity, durability, and range while maintaining fully autonomous functionality. In a full-scale implementation, payloads would include essential supplies such as medical kits, water, or survival materials. The autoloading mechanism can be scaled by enlarging the magazine to accommodate heavier payloads while preserving sequential, jam-free delivery. Structural components would be upgraded from PLA to high-strength materials such as carbon fibre or aerospace-grade aluminum to improve durability and reduce weight. The elastic launch system would be enhanced using higher-capacity torsion springs or reinforced tension elements to achieve greater launch distances while still satisfying the requirement to use stored elastic energy. Additionally, navigation and targeting would be improved through GPS and LiDAR integration to enable autonomous operation in complex, marker-less environments. Overall, the full-scale concept preserves the core three-actuator architecture validated by the prototype while scaling performance, materials, and control systems to meet real-world search-and-rescue requirements, all while remaining compliant with the original RFP constraints.

2.3 Prototype Performance

The performance of the Turbo Avenger prototype was validated through iterative trials on the simulated search-and-rescue course. The system was required to deliver three payloads into a 30 × 40 cm target area from a distance of 3 meters without human intervention. Testing demonstrated that the prototype met the required mission parameters under controlled conditions across multiple trials. The system achieved a trajectory accuracy of approximately ±5 cm, indicating consistent and repeatable launch behavior. The primary strength of the design is the high consistency of the synchronized three-actuator system. The servo-driven pulley maintains a fixed tensioning position, reducing variability between launches, while the dual-servo escapement operated reliably with no observed jams or loading failures during testing. A key limitation is the dependency on precise software timing for synchronization between the pulley release, latch mechanism, and autoloader cycle. Minor timing discrepancies in the control logic can affect launch consistency, requiring careful calibration and debugging to ensure reliable operation. Overall, the prototype met all functional requirements defined in the RFP during controlled testing.

2.4 Materials and Construction Methods

The mechanical components of the Turbo Avenger were modeled using SolidWorks and AutoCAD to ensure precise tolerances and proper integration of moving parts. Primary structural elements, including the pivoting

arm and vertical magazine, were fabricated using Red Polylactic Acid (PLA) via 3D printing. This method was selected for its ability to produce complex geometries, support rapid prototyping, and maintain sufficient rigidity under elastic loading while minimizing weight. The system employs non-destructive mounting, using standardized stainless-steel fasteners to attach the 3D-printed base plate to the existing mounting points of the Arduino ELEGOO UNO R3 chassis. This approach preserves the integrity of the platform and allows for component reuse and modular assembly. Step-by-step assembly procedures are provided in Appendix A.

The selected materials and their functional roles are summarized in **Table 2**.

Table 2:

Primary Prototype Materials and Functional Applications

<u>Material:</u>	<u>Application:</u>
PLA	Pivot arm, base plate, magazine, latch
Rubber	Elastic bands and guides
Steel	Fasteners, axle, structural support
Nylon	Pulley tension string
Electronics	Arduino Uno, ESP32, motor driver, servos

3.0 Seventh Generation Principle Reflection

The Seventh Generation Principle originates from Haudenosaunee (Iroquois) philosophy and emphasizes that decisions made today should contribute to the well-being of future generations over a long-term horizon (EcoResolution, 2025). In engineering design, this principle is applied through considerations such as sustainability, safety, resource efficiency, modularity, and minimizing environmental impact across the system lifecycle (Murphy et al., 2011). The Turbo Avenger integrates these considerations by aligning its mechanical design, material selection, and system architecture with both present operational requirements and long-term sustainability objectives.

3.1 Historical Context (7 Generations Back)

Approximately 150 years ago, engineering practice during the industrial era prioritized mechanical functionality, productivity, and rapid technological advancement, often with limited attention to long-term environmental consequences or system safety. Mechanical systems from this period typically lacked modern control systems, standardized safety mechanisms, and fail-safe designs, which increased the likelihood of uncontrolled operation and resource inefficiency. This historical context highlights the evolution of engineering toward systems that emphasize reliability and safety. In response, the Turbo Avenger incorporates a locking servo latch mechanism to ensure controlled release of stored elastic energy. This design prevents unintended launches and reflects modern engineering standards for fail-safe operation. The inclusion of controlled actuation mechanisms reduces risks associated with mechanical energy storage systems and improves overall system reliability, aligning with lessons learned from earlier generations of mechanical design (Smith & Jones, 2022).

3.2 Present Reflection

The current design of the Turbo Avenger reflects contemporary engineering priorities, including efficiency, modularity, reliability, and responsible material usage. A key feature supporting these priorities is the use of a non-destructive mounting approach, allowing the system to be integrated with the ELEGOO chassis without permanent modification. This preserves the base platform for reuse and reduces material waste, supporting sustainable prototyping practices. In addition, the use of additive manufacturing (3D printing with PLA) enables rapid prototyping while minimizing fabrication waste compared to subtractive manufacturing methods. Although PLA is used in this prototype phase due to accessibility and ease of fabrication, its selection is consistent with iterative engineering practices that prioritize development efficiency and controlled material usage. The modular nature of the system also allows individual components to be replaced or upgraded without redesigning the entire assembly, further extending system usability and reducing long-term resource consumption (Lee & Patel, 2019).

3.3 Future Impact (7 Generations Forward)

Looking forward, the Turbo Avenger design is structured to support long-term adaptability, sustainability, and responsible technological development. Future iterations may incorporate biodegradable or recycled composite materials to reduce persistent environmental waste after the system's lifecycle. Transitioning to more sustainable materials would mitigate the accumulation of non-degradable plastics and align the design with environmentally responsible engineering practices. Additionally, integrating renewable energy systems, such as solar-assisted charging, could reduce reliance on non-renewable energy sources and improve operational sustainability in remote or disaster-response environments. From a control systems perspective, upgrading the platform with sensor-based feedback mechanisms, such as ultrasonic or LiDAR sensing, would enhance adaptability in dynamic environments and reduce the need for physical redesigns. The system's scalable architecture ensures that improvements can be implemented incrementally without replacing the entire design. This adaptability supports long-term usability and reduces lifecycle waste. Overall, the Turbo Avenger aligns with the Seventh Generation Principle by emphasizing safety, modularity, material responsibility, and forward-looking adaptability, ensuring that the design remains beneficial not only for current applications but also for future generations (Murphy et al., 2011).

4.0 Product Cost

The total financial investment required for the fabrication of the Turbo Avenger autonomous delivery prototype is \$328.00 CAD. This cost includes the mobile navigation platform, the mechatronic actuation system, control electronics, and all structural components required for operation. A detailed breakdown of expenses, including component descriptions and procurement sources, is provided in Table 1.

Table 1:

Detailed Prototype Fabrication Cost Breakdown

Component	Description	Cost (CAD)
ELEGOO Smart Robot Kit	Mobile base with Arduino UNO R3 (ELEGOO, 2026)	\$100.00
Custom 3D Printing	Red PLA material and fabrication	\$141.00
Actuation System	3× micro servo motors (Amazon, 2026)	\$13.00
Control Electronics	ESP32, DRV8833 motor driver, breadboard	\$36.00
Assembly Hardware	Stainless steel fasteners kit	\$38.00
Total Prototype Cost		\$328.00

Note: Costs are based on current Canadian retail estimates and prototyping expenditures.

4.1 Cost Discussion

The total prototype cost of \$328.00 CAD is considered reasonable for a functional autonomous system with integrated actuation, control, and mobility. The design leverages commercially available off-the-shelf components, which reduces development time and eliminates the need for expensive custom electronics or machined parts. The ELEGOO Smart Robot Kit provides a reliable and standardized mobile platform, allowing the team to focus resources on developing the payload delivery and actuation subsystems. Similarly, the use of 3D printing enables rapid iteration and customization of mechanical components while maintaining acceptable strength and precision for prototype validation. Although additive manufacturing represents a significant portion of the total cost, it is justified by the need for design flexibility and geometric complexity in the synchronized three-actuator mechanism. Overall, the cost reflects a balance between affordability, functionality, and scalability. The prototype demonstrates a cost-efficient engineering solution that can be further optimized for mass production through material substitution, component integration, and manufacturing process refinement.

5.0 Conclusions and Recommendations

5.1 Synthesis of Findings and Final Performance

The Turbo Avenger successfully demonstrates that a synchronized three-actuator system can achieve reliable autonomous payload delivery for search-and-rescue applications. Testing confirmed that the prototype met all primary requirements, consistently achieving a launch range of approximately 3 meters with a trajectory accuracy of ± 5 cm while delivering three payloads within the designated 30×40 cm target area. The dual-servo electronic escapement significantly improved reliability by preventing double-feeds and eliminating jamming, enabling precise and repeatable payload release. The system also operated fully autonomously while remaining below the 1.5-meter height constraint, confirming that controlled elastic energy combined with coordinated actuation can meet both performance and safety requirements.

5.2 Technical Limitations and Lessons Learned

Testing identified software timing and actuator synchronization as the primary limitation. The system depends on tightly coordinated signals between the ESP32 and Arduino Uno to control the arming pulley, locking latch, and autoloader. Minor delays in execution can disrupt actuator sequencing, leading to inconsistent loading or firing. This highlights that system performance is constrained not only by mechanical design but also by control logic precision. While the hardware is structurally sound, the absence of closed-loop feedback reduces adaptability to dynamic conditions. The key lesson is that reliable mechanical systems must be supported by equally robust, real-time control to ensure consistent operation.

5.3 Recommendations for Implementation

Future improvements should focus on integrating real-time sensor feedback, such as ultrasonic or infrared sensing, to enable adaptive control of firing distance and improve responsiveness in changing environments. Upgrading the navigation system to incorporate GPS and LiDAR would allow operation in unstructured, marker-less environments. From a materials perspective, replacing PLA with biodegradable or recycled composites would reduce environmental impact and align the design with sustainability goals. Additionally, consolidating control into a more integrated microcontroller architecture would reduce communication delays and improve actuator synchronization. These enhancements would improve reliability, scalability, and sustainability, supporting deployment of the Turbo Avenger as a practical autonomous search-and-rescue system.

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Appendix A - Prototype Fabrication Instructions

How to Assemble and Fabricate the Turbo Avenger Prototype

1. Introduction

This manual provides a clear, step-by-step procedure for the assembly of the Turbo Avenger autonomous delivery system.

Target Audience: Engineering technicians or students familiar with basic hand tools and the Arduino IDE

Estimated Completion Time: 45–60 minutes

Required Equipment: Hex key set (2 mm, 2.5 mm), small Phillips screwdriver, needle-nose pliers, and a computer with Arduino IDE installed

2. Materials and Components List

- (1) ELEGOO Smart Robot Car Kit base
- (1) Custom 3D-Printed Base Plate (PLA)
- (1) Pivoting Launch Arm with payload cup
- (1) Vertical Magazine Tube
- (3) Micro Servos (SG90 or equivalent)
- (1) DC Motor with pulley attachment
- (4) High-tension elastic bands
- (1) Stainless steel bolt and nut kit

3. Assembly Procedure

Phase 1: Structural Mounting

1. Clear the top acrylic plate of the ELEGOO robot car to prevent obstruction of sensors and wheels.
2. Align the custom 3D-printed base plate with the existing mounting holes on the chassis, as shown in Figure A1.

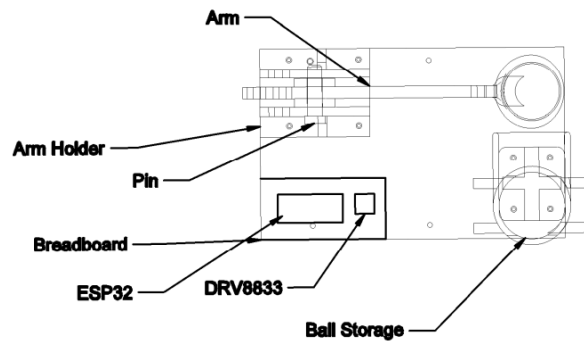


Figure A1: Orthographic view of the Turbo Avenger base plate alignment on the ELEGOO chassis.

3. Secure the base plate using four M3 bolts and nuts.

Note: Do not over-tighten fasteners, as PLA may crack under excessive stress.

Phase 2: Actuator Installation

4. Insert the locking servo into the designated rear slot on the base plate.
5. Attach the dual-servo autoloader to the vertical magazine tube using mounting brackets.
6. Mount the DC motor to the side frame to function as the arming pulley.

Phase 3: Launch Mechanism

7. Install the pivoting launch arm into the central axle support and secure it with the main pivot bolt.
8. Loop four high-tension elastic bands over the tensioning points on the arm and base.

Warning: Keep hands clear of the launch path during this step to prevent injury.

9. Thread the arming string from the motor pulley to the rear of the launch arm.

Phase 4: Electronics and Wiring

10. Connect the three servo signal wires to the ESP32 control interface.
11. Attach the DC motor leads to the DRV8833 motor driver.
12. Link the Arduino control pin to the ESP32 input pin to enable synchronized operation, as shown in **Figure A2**.

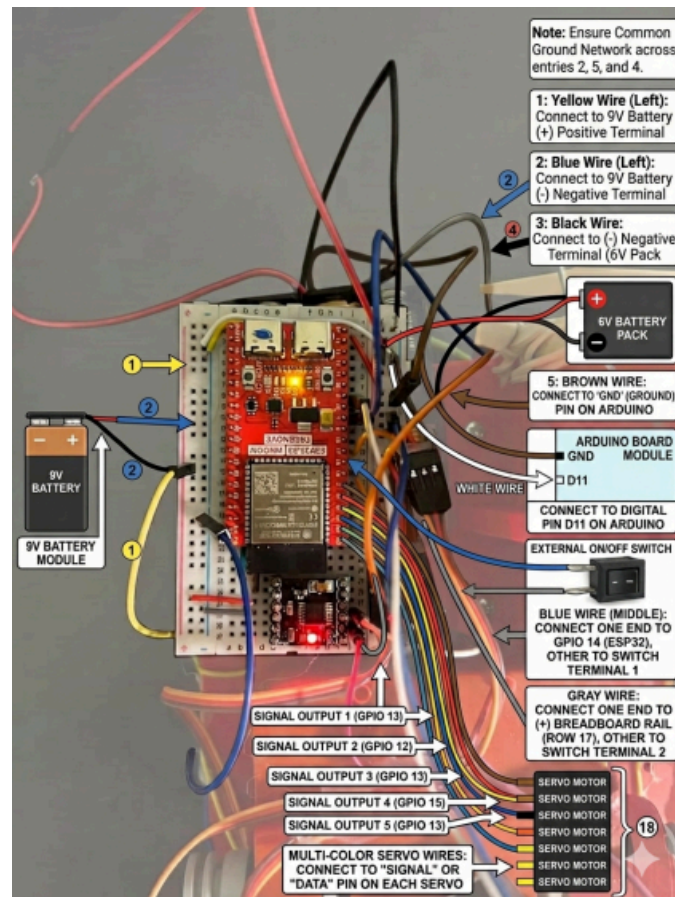


Figure A2: Actuator wiring configuration showing servo connections, ESP32 interface, and DRV8833 motor driver.

4. Final Check and Operation

13. Perform a manual dry run by rotating the pulley to confirm the arm engages the locking latch correctly.
14. Verify that all fasteners and connections are secure.
15. Upload the control program and initiate autonomous testing.

Conclusion

The assembly of the Turbo Avenger is now complete. Perform a manual "Dry Run" by rotating the pulley to ensure the arm catches the locking latch without obstruction before applying battery power. Once verified, upload the control code to initiate autonomous trials.

Appendix B - AutoCAD Drawing(s) of Design

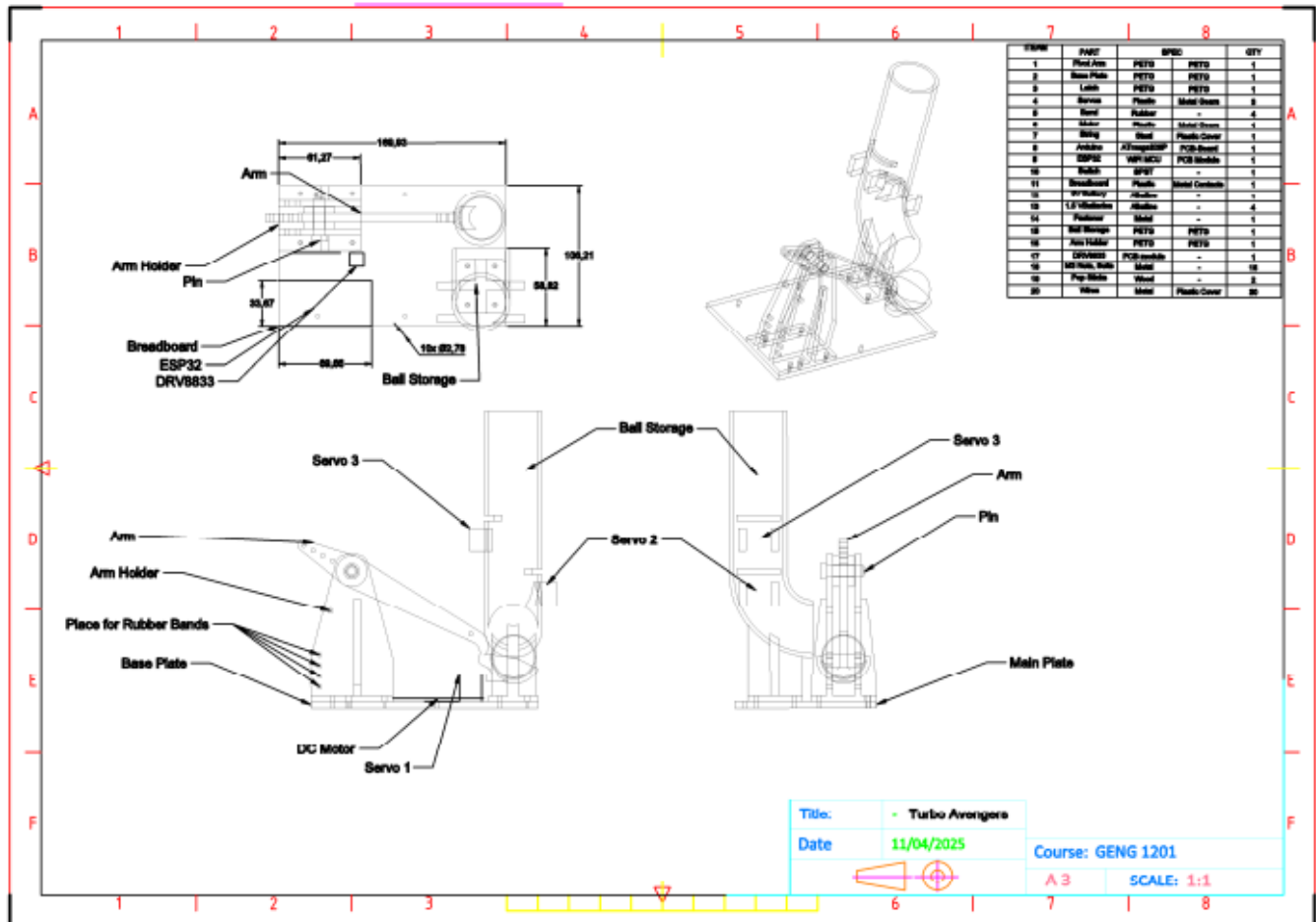


Figure B1: Fully dimensioned engineering drawing set of the Turbo Avenger prototype, including front, top, and right-side orthographic views and a 3D isometric model, illustrating component geometry, system integration, and compliance with chassis dimensional constraints.

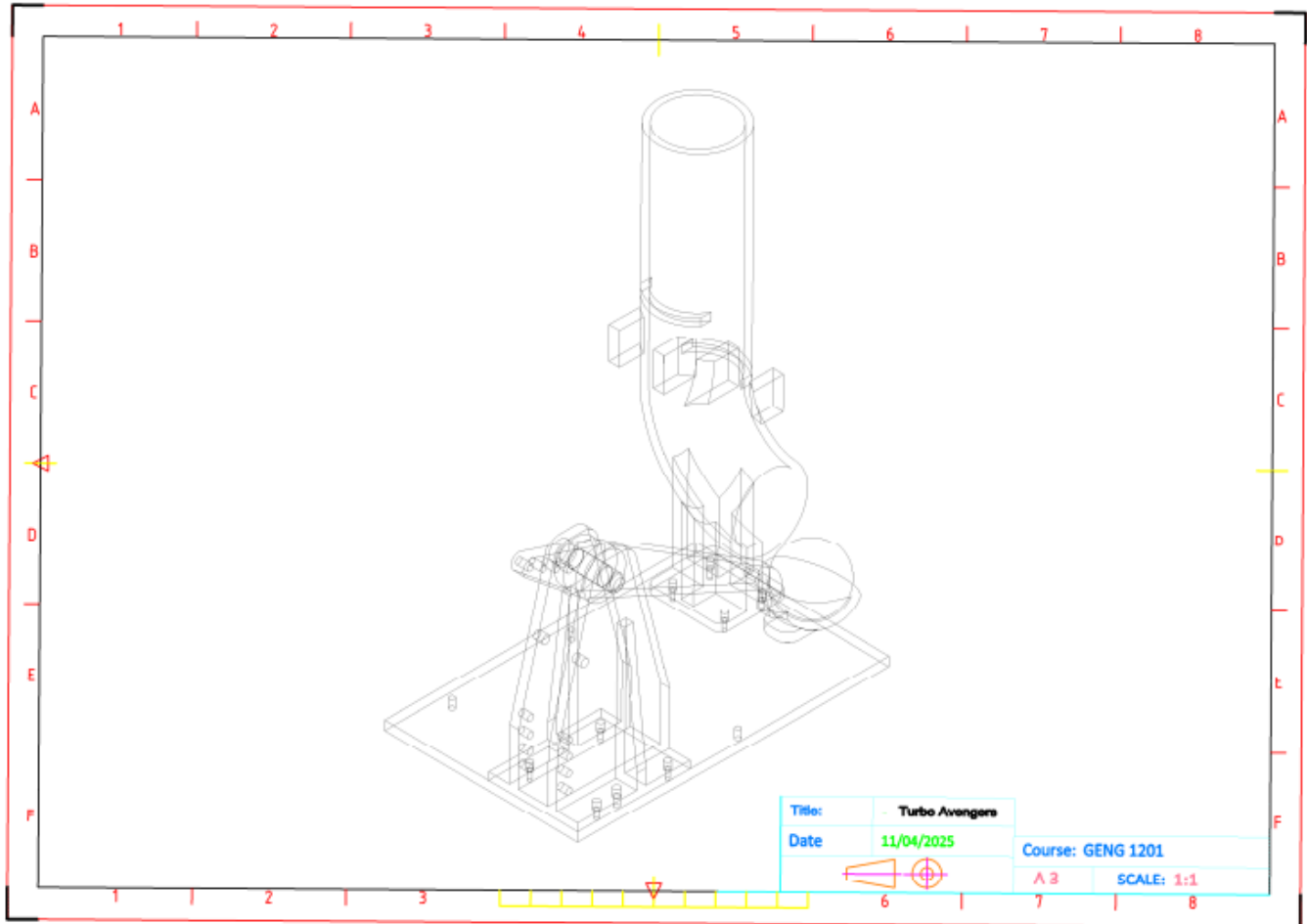


Figure B2: 3D isometric concept rendering of the Turbo Avenger system.

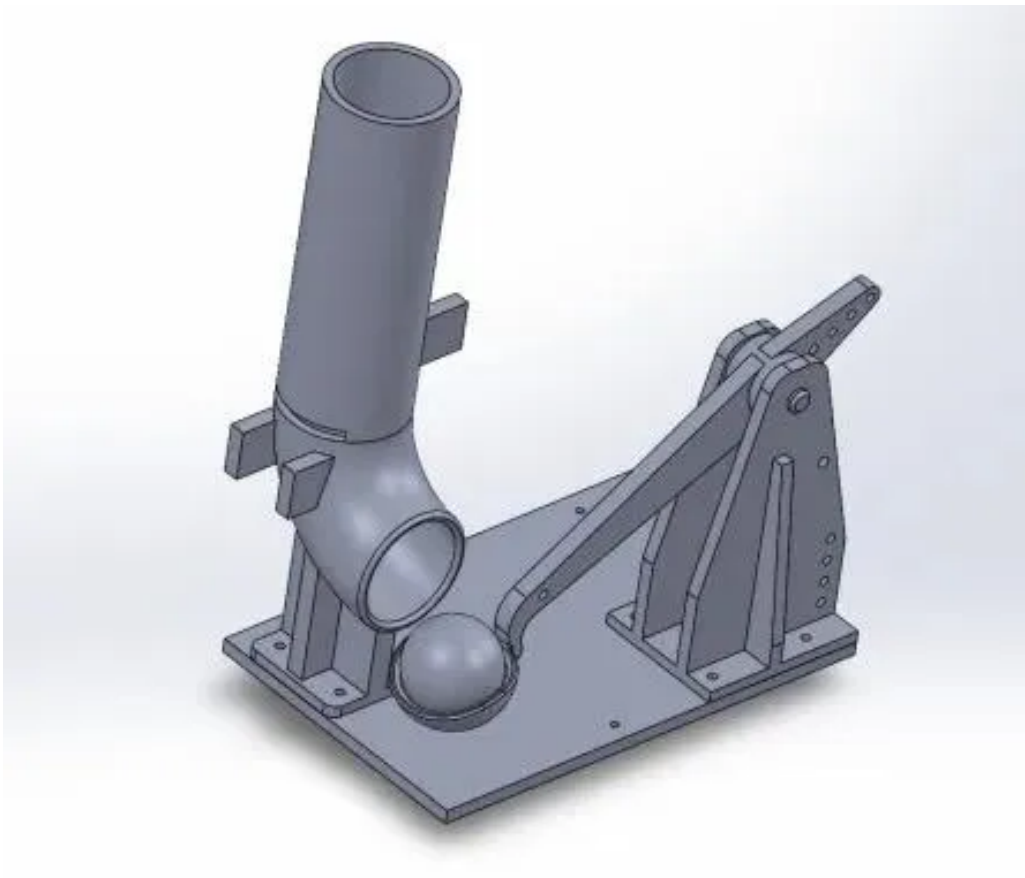


Figure B3: 3D AutoCad Model of the Turbo Avenger system.

Appendix C – Prototype Cost Breakdown Table

Table C1: Detailed Prototype Fabrication Cost Breakdown

The following table provides a comprehensive itemization of the materials and components utilized in the construction of the Turbo Avenger prototype. All costs are listed in Canadian Dollars (CAD) and reflect verified retail pricing and fabrication estimates.

Table C1:

Detailed Prototype Fabrication Cost Breakdown

Component	Technical Description & Source	Cost (CAD)
Mobile Base	ELEGOO Smart Robot Car Kit V4 (Arduino UNO R3 platform) (ELEGOO, 2026)	\$100.00
Custom Structure	Red PLA 3D-printed components (pivot arm, base plate, magazine) (University of Windsor, 2026)	\$141.00
Actuation System	3× SG90 micro servo motors (Amazon, 2026)	\$13.00
Control Mechatronics	ESP32 WROOM module, DRV8833 motor driver, breadboard (Electronic Supply, 2026)	\$36.00
Mechanical Fasteners	Stainless steel M3/M4 bolt and nut kit (Home Hardware, 2026)	\$38.00
Energy Storage	High-tension elastic bands (existing inventory) (Team Inventory, 2026)	\$0.00*
TOTAL COST		\$328.00

Note. All pricing is based on Canadian retail estimates and university fabrication data. Items listed at \$0.00 were sourced from existing inventory and are excluded from total procurement cost.